

Chapter 5

Cleaning Validation and Its Regulatory Aspects in the Pharmaceutical Industry

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1 INTRODUCTION

In recent years, cleaning has achieved a position of increasing importance in the pharmaceutical industry. Virtually every aspect of manufacturing involves cleaning, from the initial stages of bulk production to the final dosage form to ensure the safety, identity, strength, quality, or purity of the drug product. Modern pharmaceutical manufacturing involves highly technical personnel, complex equipment, sophisticated facilities, and complicated processes.

The purpose of cleaning validation is to prevent contamination and cross-contamination in pharmaceutical dosage forms. The pharmaceutical products' manufacturer must reproduce consistently the desired quality of products. The current good manufacturing practice regulation recognizes that contamination and cross-contamination of pharmaceutical products must not occur. The control of cross-contamination plays a very important role in maintaining the quality of the product [1–3].

Cleaning validation and verification is routinely conducted on processing equipment. This requirement is very important in clinical functions. Development of a cleaning validation system is a rigorous, tedious, and time-consuming process. Cleaning validation is necessary for the following reasons:

1. It is a prime customer requirement since it ensures the purity and safety of the product.

2. It is a regulatory requirement in active pharmaceutical ingredient (API) product manufacture.
3. It also assures the quality of the process through internal control and compliance.

Inadequate cleaning procedures result in any number of contaminants that may be present in the next batch manufactured using the same equipment, such as

1. precursors to the API;
2. by-products and/or degradation products of the API;
3. the previous product;
4. solvents and other materials employed during the manufacturing process;
5. microorganisms particularly where microbial growth may be sustained by the product; and
6. cleaning agents themselves and lubricants.

1.1 Purpose

When a single equipment facility is used for multiple products, cleaning validation provides the following information:

- Requirements for cleaning plant and equipment used to manufacture APIs or their intermediates.
- How to assure appropriate cleaning of API plants and equipment.
- When is validation applicable and what must be done to complete the validation [4–7].

API: Any substance or mixture of substances intended to be used in the manufacture of a pharmaceutical dosage form and that, when used as intended, becomes an active ingredient of that pharmaceutical dosage form. Such substances are intended to furnish pharmacological activity or other direct effect in the diagnosis, cure, mitigation, treatment, or prevention of disease, or to affect the structure and function of the body [8].

1.2 Reasons

There are several reasons for validation of the cleaning process:

- Initial qualification of a process/equipment
- Critical change in a cleaning procedure
- Critical change in formulation
- Significant change in equipment
- Change in cleaning process
- Change in cleaning agent [9].

1.3 Contaminant

The reagent or intermediate is generally the contaminant within the equipment that is considered to have the greatest impact on patient safety. This is assessed by taking into consideration information on toxicity, ease of removal, and pharmacological activity.

There may be a need to define more than one contaminant if there is more than one potential contaminant with significant toxicological/pharmacological activity that is removed by a different cleaning method. For example, a heavy metal from a catalyst may be considered as a guiding substance along with the final product from the equipment train. Contamination of the product due to the presence of residue on the surface of the cleaned equipment is shown in Fig. 5.1. Formation of new API due to the presence of residue on the surface of the cleaned equipment is shown in Fig. 5.2.

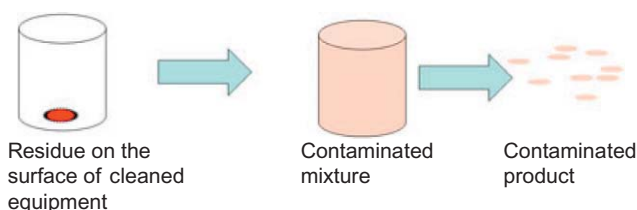


FIGURE 5.1 Contamination due to presence of residue of the previous product.

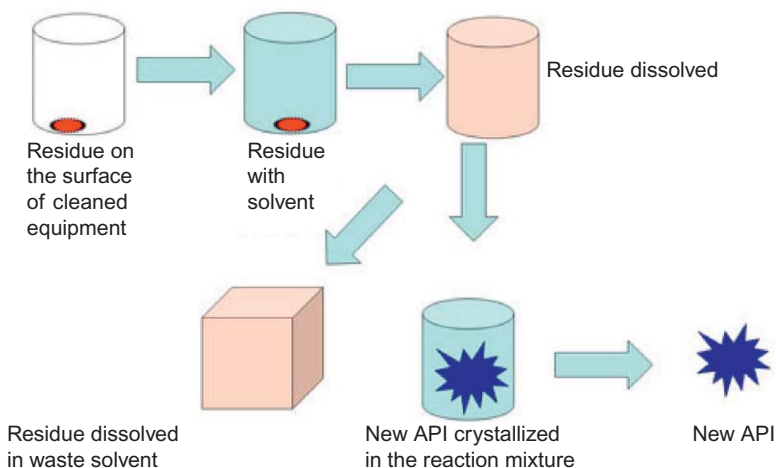


FIGURE 5.2 Formation of new API due to contaminant.

1.3.1 *Types of Contamination*

1. Cross-contamination with active ingredients. Contamination of one batch of product with significant levels of residual active ingredients from a previous batch cannot be tolerated. In addition to the obvious problems posed by subjecting consumers or patients to unintended contaminants, potentially clinically significant synergistic interactions between pharmacologically active chemicals are a real concern.
2. Contamination with unintended materials or compounds. While inert ingredients used in drug products are generally recognized as safe or have been shown to be safe for human consumption, the routine use, maintenance, and cleaning of equipments provide potential sources of contaminants, including equipment parts, lubricants, chemical cleaning agents, and pieces of cleaning items such as brushes and rags.
3. Microbiological contamination. Maintenance, cleaning, and storage conditions may provide adventitious microorganisms with the opportunity to proliferate within the processing equipment [10,11].

2 GOOD MANUFACTURING PRACTICE IN API MANUFACTURING

Good manufacturing practice in API manufacturing proceeds from initial process steps to final process steps. Similarly, cleaning the equipment proceeds from the initial cleaning step to the final cleaning step, which are reflected in the quality of the API and subsequently in the drug product based on effective cleaning and validation.

2.1 **Designing the Cleaning Process in Manufacturing Plants**

Manufacturing of pharmaceutical products involves a series of processing steps and use of various equipment. Equipment and ancillary systems are used for manufacturing of multiple products or a single dedicated product. Inadequate cleaning processes may carry forward the residue as a contaminant in the next batch to be manufactured in the same equipment [3,10,12,13].

2.2 **Equipment in the Manufacturing Plant**

Several aspects should be considered for the equipment in the manufacturing plant.

2.2.1 *Equipment Design Considerations*

Good design for cleaning should be built into plant/equipment specifications. Equipment cleaning performance and function must be considered during equipment design. Ideally, the equipment should be constructed of nonreactive, nonadditive, nonadsorptive, and nonporous materials. The equipment should

have smooth surfaces; it should be free from pitting; it should be free draining; it should have limited intricate or complex parts; and it should not be additive or absorptive. Cleaning equipment should be designed to ensure access to all process equipment surfaces to be cleaned.

The first activity is to ensure the cleanliness in manufacturing plants by actively considering the plant/equipment design. Equipment cleaning aids, such as bristles, brushes, and cleaning cloths made of shedding materials, that raise dust or generate contamination should not be used. The manufacturer must ensure that all materials from the previously manufactured product are removed, and after cleaning, there should be no residual cleaning agent left behind. Repair and maintenance operations should not present any hazard to the quality of the product [14].

2.2.2 *Equipment Characteristics*

Equipment usage during production is another important aspect to consider in establishing a cleaning validation program. It is important to understand the range of products that are likely to come in contact with the various equipment surfaces; this will help to identify the contamination and cross-contamination potential of the equipment.

2.2.3 *Construction Materials*

Equipment should be constructed such that surfaces which contact components, in-process materials, or drug products will not be reactive, additive, or absorptive, and not alter the safety, identity, strength, quality, or purity of the drug product. Any substances that are required for operation, such as lubricants or coolants, should not come into contact with the components, drug product containers, closures, in-process materials, or drug products, as this could alter the safety, identity, strength, quality, or purity of the drug product.

2.2.4 *Dedicated and Nondedicated Manufacturing Equipment*

Dedicated equipment is used solely for the production of a single product which will markedly reduce cross-contamination. When the same piece of equipment is utilized for a range of product formulations, the prevention of cross-contamination between products becomes the main objective in the cleaning validation effort.

For new plants, a list of potential hot spots should be included in the documentation. Experimental cleaning procedures may be conducted prior to production to identify the hot spots, and how easily the equipments can be cleaned by specific cleaning procedures in order to collect data to establish optimal cleaning procedures.

Only dedicated production areas should be considered when a material of high pharmacological activity or toxicity is involved (e.g., penicillin, certain

steroids, or cytotoxic anticancer agents) unless validated inactivation and/or cleaning procedures are established and maintained [2,15,16].

2.3 Personnel

All employees should undergo health examination. Personnel should wear clean, appropriate body covering during manufacturing, and direct contact between the operator's hands, and starting materials, primary packing materials, and intermediate and finished products should be avoided.

2.4 Heating, Ventilation, and Air Conditioning System

One of the most important areas in pharmaceutical process control is the development of systems to control microorganisms during manufacturing and storage of equipment. Install appropriate dedicated and validated heating, ventilation and air conditioning (HVAC) system in all manufacturing areas with suitable air locks and pressure differentials. Minimize the risk of contaminants caused by recirculation/reentry of untreated/insufficiently treated air. Maintain a separate area for cleaning of filters away from the air handling unit.

2.5 Clothing and Footwear

Employees should minimize exposure of body surfaces during manufacturing. The operator should ensure that clothes are clean during product changeover and are changed after being contaminated by cleaning agents. The manufacturer should provide appropriate footwear to the personnel operating in particular areas and should not permit personal footwear inside the manufacturing area [3,17,18].

3 ESTABLISHING THE ACCEPTANCE LIMITS

The determination of cleaning limits and acceptance criteria is a crucial element of a good cleaning validation program. Establishing the limits for product residues should be logically based on consideration of the materials involved in the manufacturing and their therapeutic dose. The limits and acceptance criteria should be practical, verifiable, achievable, and scientifically sound.

Establishing the limits can be based on:

- Product-specific cleaning validation for all products.
- Grouping into product families and choosing a worst-case product.
- Grouping by groups of risk (e.g., very soluble products, similar potency, highly toxic products, or difficult to detect).
- Setting limits on not allowing more than a certain fraction of carryover.
- Different safety factors for different dosage forms.

3.1 Doses

Maximum daily dose (MDD)

The maximum dose of active substance {usually milligrams (mg) or grams (g)} typically administered to a patient in any 24-h period.

Minimal effect dose (MED)

The minimum dose at which there is an observable pharmacological effect in a human being. The MED is expressed as a weight of active substance (usually mg or g) per day.

Minimum therapeutic dose (MTD)

The minimum amount of active substance (usually mg or g) typically administered to a patient on each occasion.

3.2 Approaches in Establishing the Acceptable Carryover Quantity

Several approaches are used to establish acceptable carryover quantities:

1. Dose criterion. No more than 0.1% of the normal therapeutic dose of any product will appear in the MDD of the following product.
2. 10 ppm criterion. No more than 10 ppm of any product will appear in another product.
3. Visually clean criterion. No residue should be visible on the equipment after cleaning procedures are performed.

The acceptable carryover quantity is determined to ensure that the level of residue after cleaning will not have a clinically significant pharmacological or toxicological effect at the MDD of the subsequent product. Typically, the contaminant might be the last material prepared in the vessel, although other components of the contamination matrix, such as catalysts, toxic reagents, solvents, degradedness, or by-products of the last material, could also be the reason for contamination of the product.

The amount of a specific contaminant actually present in the equipment is determined by summing the amounts present in the rinse washes or swabs of all equipments used in manufacture. For the calculation a worst-case assumption is made that the amount of contaminant remaining is equal to the amount that has been recovered by swab sampling or from rinse analysis.

Acceptance limits for pharmaceutical manufacturing operation are described in Sections 3.2.1 to 3.2.3 [2,14,19–22].

3.2.1 Approach 1 (Dose Criterion)

No more than 0.001 of minimum daily dose of any product will appear in the MDD of another product.

Milligrams of active ingredient in product A permitted per 4 in.²

$$\text{swab area} = \frac{I \times K \times M}{J \times L}$$

I = 0.001 of the smallest strength of product A manufactured per day expressed as mg/day and based on the number of milligrams of active ingredient.

J = Maximum number of dosage units of product B per day

K = Number of dosage units per batch of final mixture of product B

L = Equipment surface area common between products A and B expressed as square inches

M = 4 in.²/swab.

3.2.2 Approach 2 (10 ppm Criterion)

Any active ingredient can be present in a subsequently manufactured product at a maximum level of 10 ppm.

Milligrams of active ingredient in product A permitted

$$\text{per 4 square inches (in.²) swab area} = \frac{R \times S \times U}{T}$$

R = 10 mg active ingredient of product A in 1 kg of product B

S = Number of kilograms per batch of final mixture of product B

T = Equipment surface area common between products A and B expressed as square inches

U = 4 in.²/swab.

3.2.3 Approach 3 (Visually Clean Criterion)

No residue should be visible on the equipment after cleaning procedures are performed.

3.3 Limits Based on Medical or Pharmacological Potency of the Product

One basis of establishing limits is a mathematical calculation which allows a certain fraction of the therapeutic dose to carry over into each dosage unit of the following product. The safety factor approach for establishing the carryover limits is shown in [Table 5.1](#).

3.3.1 The Basis for Quantitative Limits

Actual numerical limits are usually based on one of the following:

- The medical or pharmacological potency of the product
- The toxicity of the residue

TABLE 5.1 Safety Factor Approach for Establishing the Carryover Limits

Approach	Approach Typically Applicable to
1/10th to 1/100th of a normal daily dose	Topical products
1/100th to 1/1000th of a normal daily dose	Oral products
1/1000th to 1/10,000th of a normal daily dose	Injections, ophthalmic products
1/10,000th to 1/100,000th of a normal daily dose	Research, investigational Products

- The analytical limit of detection

$$\text{MACO} = \frac{\text{TD} \times \text{BS} \times \text{SF}}{\text{LDD}}$$

MACO=maximum allowable carry over
TD=single therapeutic dose
BS=batch size of the next product to be manufactured in the same equipment
SF=safety factor
LDD=largest daily dose of the next product to be manufactured in the same equipment.

3.4 Limits Based on Toxicity of the Residue

Using the therapeutic dose as the basis for calculations of limits is appropriate for situations where the material is an active ingredient and therapeutic dosage levels are known. However, there are other situations where the material is not medically used and no therapeutic dose data are available.

When toxicity is expressed as LD₅₀, the following methodology can be used [2,14,19–22]:

$$\begin{aligned}\text{NOEL} &= \text{LD}_{50} \times \text{EF} \\ \text{ADI} &= \text{NOEL} \times \text{AAW} \times \text{SF}\end{aligned}$$

NOEL=no observed effect level
LD₅₀=lethal dose for 50% of animal population in the study
EF=empirical factor (derived from an animal model)
ADI=acceptable daily intake
AAW=average adult weight
SF=safety factor

$$\text{MACO} = \frac{\text{ADI} \times B}{R}$$

B = smallest batch size of any subsequent product

R = largest daily dose of any product made in the same equipment.

3.5 Risk Levels in Cleaning Validation

Three risk levels are considered depending on the type of changeover in manufacturing [23]. The considerations are given in the table below:

Route of Administration	Dose Level	Acceptance Criteria
API administered by inhalation/ parenterally	API nil effect dose (NED)	10
	API minimum effect dose (MED)	10
	API MTD	100

If the contaminants are metals, the concentration limits (ppm) for metals are calculated based on the permitted daily exposure limits listed in the table below:

Elements	Oral Concentration Limits (ppm)	Parenteral Concentration Limits (ppm)
Pt, Pd, Ir, Rh, Ru, Os	5	0.5
Mo, V, Ni, Cr	15	1.5
Zn, Fe	20	2.0

Calculation for ACQ limit of metals is based on the following equation:

$$\text{ACQ}_{\text{metal}} = \frac{\text{CL} \times \text{MBS}_{\text{API}}}{1,000,000}$$

MBS_{API} = Minimum batch size of API (mg)

$\text{ACQ}_{\text{metal}}$ = Acceptable carryover quantity (mg) into API

CL = Concentration limit (ppm) from table above

When the calculated ACQ level using this methodology yields values that are not practically achievable, an alternative, higher ACQ may be justified and ratified by the cognizant drug safety operations review committee:

$$\text{ACQ} = \frac{\text{Toxicity value} \times \text{MBS}_{\text{API}}}{\text{RAF} \times \text{MDD}_{\text{API}}}$$

Toxicity value is equal to NOEL, where available. If NOEL is unavailable, the following parameters may be used for the calculation of ACQ:

ACQ = Acceptable carryover quantity (mg) into API

MBS_{API} = Minimum batch size of APIb (mg)

MDD_{API} = Maximum daily dose of API (mg)

3.6 Use of ACQ for a Level 0 or Level 1 Changeover

Generally, it is not possible to use the calculation approach for level 0 or 1 changeover because therapeutic dose information is not available. However, in some circumstances, a calculated ACQ, or, where this is not possible, a more stringent ACQ may be applied to level 0 or 1 changeovers. The risk level minimum acceptance criteria is shown in [Table 5.2](#).

4 CLEANING OF EQUIPMENT

The cleaning processes used in industry rely upon solubilization, chemical reaction, and physical action for residue removal.

Key factors in cleaning validation

- Selection of equipments (based on worst-case approach).
- Appropriate solvent/detergent (based on solubility data).
- Cleaning procedure (hand scrubbing/solvent wash/clean in place (CIP)/clean out of place/quantities/time/pressure/temperature).
- Level of cleaning required (based on risk assessment).
- What is clean (acceptance criteria based on visually clean/rinse limit/swab limit/microbiological aspects).

4.1 Cleaning Methods

Many methods are available for cleaning.

4.1.1 Types of Residue

When establishing a cleaning program, it is important to first identify the substance to be cleaned. The physical and chemical properties of the residue need to be considered that can affect the cleaning process. Degree of solubility, hydrophobicity, reactivity, and other properties can affect the characteristics of these residues during the cleaning process. Cleaning methods must reflect actual equipment usage for calculating the predetermined acceptance criterion. Typical cleaning methods include water rinses, acid/base rinses, refluxing solvent washes, high-pressure water jets, manual cleaning, rinsing via spray ball, solvent recirculation (via filtration), and detergent cleaning.

4.2 Cleaning Mechanisms

Cleaning can be defined as the removal of residues left behind from a previous batch, other residues, and traces of cleaning agents. There are several mechanisms associated with cleaning of equipment.

The mechanisms involved can be mechanical action and chemical action between the residues and the cleaning agent. The selection of cleaning agent and mechanism involved in cleaning is largely dependent on the process residue to be cleaned.

TABLE 5.2 Risk Levels in Cleaning Validation

Risk Level	Risk Consideration	Minimum Acceptance Criteria for the Equipment Train
0	Carryover of material within the same synthesis sequence represents the lowest risk	The acceptance criteria must be based on technical considerations (impact on chemistry and API purity profile) and can be assessed through risk assessment The minimum acceptance criterion (i.e., if there are no technical restrictions on the acceptable level of carryover) is that the equipment must be free from gross contamination
1	Carryover of the intermediate/ crude API into a different synthesis sequence or into final purification step of the same synthesis sequence represents a higher risk to product quality The carryover of material into the subsequently purified API will be reduced through attrition (e.g., loss to mother liquors, screening filtrations)	Theoretically toxicological/ pharmacological data could be used to calculate the ACQ; however, the carryover to the following final API will depend on the yields of the subsequent reaction/purification steps, the relative solubility of the contaminant and the stability of the contaminant under the conditions of the succeeding process steps. Since all the information required for calculating an ACQ is unlikely to be available, a default minimum acceptance criterion of 100 ppm w/w carryover for the guiding substance is applied (e.g., 100 mg of carryover per kg of the next product)
2	Carryover of an intermediate or an API into the purification step of an API (or postfinal purification step such as milling and blending) from a different synthesis represents the highest risk because of the potential for unrelated toxicity/activity effects The likelihood of attrition is reduced as there is only one processing step (there will be some reduction in equipment in contact with solvent such as dissolution vessels and crystallizers; however, there will be none in dryers and mills)	The acceptance criterion is calculated from toxicity/activity data. If no data are available, then a default acceptance criterion of 10 ppm w/w carryover of the guiding substance is applied (e.g., 10 mg of carryover per kg of the next product) If the calculated acceptance criterion is greater than 100 ppm w/w, then a default limit of 100 ppm w/w must be applied If the calculated acceptable carryover limit is between 10 and 100 ppm, then the calculated limit must be applied

4.2.1 *Types of Cleaning Mechanisms*

The cleaning mechanism depends entirely on the selection of cleaning agent and type of residue to be cleaned. Various types of cleaning mechanisms are listed below:

1. Mechanical action
2. Dissolution
3. Saponification
4. Detergency (includes wetting agents and emulsification)
5. Chemical reaction.

Many cleaning agents perform several functions at once. For instance, butyl (glycol ether family) can serve as a wetting or surface tension reducing agent as well as a solubilizing agent. It can also contribute to emulsifying capability when combined with anionic surfactants or soaps (alkali metal salts of carboxylic acids).

4.2.1.1 Mechanical Action

This refers to the removal of residues and contaminants through physical actions such as brushing, scrubbing, and use of pressurized water.

4.2.1.2 Dissolution

Dissolution is the process by which a solid or a liquid forms a homogeneous mixture with a solvent or solution. This can be explained as a breakdown of the crystals into individual ions, atoms, or molecules and their transport into the solvent. The mechanism involved in this type of cleaning is solubility of the residue in the cleaning agent or solvent. The monobasic buffers, such as sodium chloride, are soluble in cold and hot water for injection (WFI). Ethylene glycol butyl ether is soluble in water, and oil is also used as a solubilizing agent. Chelating agents and builders are added to the formulation to keep water hardness from interfering with the cleaning process.

The rate of dissolution depends on several factors:

- Nature of solvent or residue to be dissolved
- Temperature of solvent
- Mixing
- Area of contact
- Presence of inhibitors.

4.2.1.3 Saponification

Saponification can be defined as a “hydration reaction where free hydroxide breaks the ester bonds between the fatty acids and glycerol of a triglyceride, resulting in free fatty acids and glycerol,” which are each soluble in aqueous solutions. This process specifically involves the chemical degradation of lipids,

which are not freely soluble in aqueous solutions. Heat-treated lipid residues are more difficult to remove than nonheat-treated residues due to polymerization. Saponification plays a critical role in cleaning lipids which are present in process areas involving cell growth and cell processing, such as bacterial fermentation and cell disruption process.

4.2.1.4 Detergency

This refers to the power or quality of cleansing.

4.2.1.4.1 Wetting Wetting can be defined as a process which “involves the lowering of the surface tension of the cleaning solution, thus allowing it to better penetrate residues that are adhered to equipment and piping surfaces.” Wetting agents, or surfactants, are often used in relatively small amounts and they can substantially reduce the quantities of cleaning agent (in this case, alkali) required for residue removal.

Advantages of wetting

- Lowers the surface tension of the cleaning solution
- Allows better penetration of residues which are adhered to the equipment
- Used in small amounts
- Sticky residues that are hydrophobic in nature are easily removed.

Water acts as a solvent that breaks up solid particles after the surfactants reduce the surface tension and allow the water to penetrate the soil (water is commonly referred to as the universal solvent).

4.2.1.4.2 Emulsification Emulsifiers and suspension agents are often used to keep residues from precipitating by providing “hydrophobic groups” onto which hydrophobic areas of residues can associate, thus preventing them from associating with other residues and forming larger particles which are likely to leave the solution. These agents also typically have “hydrophilic groups” which keep them very soluble in aqueous solutions of moderate to high ionic concentrations. Emulsifiers, including anionic soap surfactants, cationic surfactants, and neutral surfactants, increase the capacity of a cleaning agent to emulsify nonsoluble compounds in the cleaner.

Advantages of emulsifiers

- Prevent aggregation of residues
- Allow the residue to precipitate and not allow the residue to redeposit on surface.

4.2.1.5 Chemical Reaction

Oxidation and hydrolysis reactions chemically break the organic residues and contaminants to make them readily removable from the equipment [24–26].

4.3 Grouping of Products

The products are first grouped according to formulation and dosage form, based on their potency, toxicity, and solubility. Products are further subdivided based on the kind of equipment used during manufacture. Further differences are made according to cleaning method and agent.

Once the product groups have been established, then it is possible to determine the “worst case” in each group, based on its toxicity and solubility or the presence of ingredients known to be difficult to clean.

4.4 Cleaning Process

The interaction of the product with all surfaces with which it will come into contact is critical.

Characteristics such as the solubility, concentration, physical properties of the active and excipients, possible degradation products, and the effect of the cleaning agent are critical factors in establishing the cleaning method.

The four widely accepted critical process parameters are time, temperature, cleaning agent concentration, and cleaning action (impingement, sheeting, rinsing, etc.) for reproducibility of the cleaning process. Validation of a cleaning process should be based on a worst-case scenario which includes the following:

- Challenge of the cleaning process to show that the challenge drug substance can be recovered in sufficient quantity or demonstrate log removal to ensure that the cleaning process is indeed removing the drug substance to the required level.
- The use of reduced cleaning parameters such as overloading of contaminants, overdrying of equipment surfaces, minimal concentration of cleaning agents, and/or minimum contact time of detergents.

4.4.1 Types of Cleaning Processes

Two types of cleaning processes are available.

- Manual
- Semiautomated/automated.

A comparison between manual and CIP cleaning sequences is shown in [Table 5.3](#).

In all cases, the cleaning procedure must prove to be effective, consistent, and reproducible. With the manual procedure, one must rely on the operator skills, and thorough training of the operator is necessary to avoid variability in performance. However, in some instances, it may be more practical to use only manual procedures.

The United States Food and Drug Administration (FDA) recommends CIP should be used to clean process equipment and storage vessels in order to reproduce exactly the same procedure each time.

TABLE 5.3 Comparison of Manual Cleaning Sequence with Clean-In-Place Sequence

Manual Cleaning Sequence	CIP Cleaning Sequence
Dismantle the parts of equipment to be cleaned	Prewash the parts in tap water
Prewash the parts with tap water	Wash the prewashed parts with cleaning solution
Wash the prewashed parts with cleaning solution	Blow out using compressed air
Rinse the parts in tap water	Rinse the parts with tap water
Rinse now with purified water	Final rinse using purified water
Dry the parts using hot air	Blow out using compressed air
Visual inspection is done to check whether the equipment is clean	Drying using hot and compressed air
Reassemble the parts finally	

4.5 Cleaning Porous Equipment

The cleaning of porous equipment is a concern due to the ability of the surface to absorb drug products. In this case, dedicated equipment would be the preferred choice. If dedicated porous equipment is not available, then cleaning procedures for soaking the material in a solvent or extracting the absorbed drug product may be required.

4.6 Cleaning Frequency

The frequency of cleaning is usually determined based on the nature of the next product to be manufactured. The number of times cleaning is required depends on the factors mentioned below. (However, to achieve acceptance criteria, “test until clean” is not acceptable.) This concept involves cleaning, sampling, and testing with repetition of this sequence until an acceptable residue limit is attained. The following factors need to be considered in determining the cleaning frequency:

- Interval between the end of production and the beginning of the cleaning procedures [dirty equipment hold time study].
- The period and, when appropriate, conditions of storage of equipment before cleaning and the time between cleaning and equipment reuse [clean equipment hold time study].

4.7 Product Attributes

The differences in the cleaning of equipment utilized for solid and liquid formulations are quite significant. The route of administration of a product may affect the level at which the product is found to be allergenic, toxic, or potent. Injectable products, intraocular formulations and some inhalants have direct access to systemic circulation. Hence, sterile manufacturing facilities must control microbial, endotoxin, and particulate levels in the manufacturing area. Liquid formulations may have greater ability to penetrate equipment seals and joints, hindering their removal during cleaning. Solid formulations may have unique abilities to form aggregates of product (clumping), which prevents wetting of the equipment surface by cleaning agents and limits the ability to rinse the residual product from the equipment surface.

4.7.1 *Between Batches of Different Products*

When equipment is changed over from one product to another, cleaning must take place to prevent product cross-contamination.

4.7.2 *Between Batches of the Same Product*

The cleaning frequency within batches of the same product should be determined as part of the process development.

4.8 Postcleaning Equipment Storage

After cleaning is completed, the equipment should be protected from cross-contamination due to the presence of other products or processes in the area. Wrap the cleaned equipment with polythene bags until use. This involves covering the equipment or placing it in an area that is free from possible cross-contamination and is designated for cleaned equipment. Visual inspection of the equipment immediately before use is necessary. Records must be kept showing the equipment number, the date of the cleaning, who cleaned it, and who inspected/tested it.

4.9 Microbiological Considerations

Microbiological aspects of equipment cleaning should be considered as preventive measures rather than removal of contamination by microorganisms. Routine cleaning and proper storage of equipment prevent microbial proliferation. The equipment should be dried before storage and under no circumstances stagnant water should be allowed to remain inside the equipment after cleaning. Time frames between the storage of unclean equipment and prior to beginning of cleaning, as well as time frames and conditions for the storage of cleaned equipment, should be established. Adequate cleaning and storage of equipment are important to ensure that subsequent sterilization or sanitization procedures achieve the necessary assurance of sterility.

4.10 Documentation

Depending on the nature of the cleaning process, documentation details may vary. When a complex cleaning procedure is required, it is important to document all the steps along with the information about who had cleaned it, when the cleaning was carried out, and the product which was previously processed on the equipment being cleaned.

4.11 Inspection and Sampling Plan

An effective inspection and sampling plan must be developed and documented based on plant layout and equipment with particular attention to hot spots. A hot spot is defined as a surface that is judged to be hard to clean or has the potential to obstruct the materials. Critical hot spots of equipment are assessed using a block diagram of the equipment by a team consisting of quality assurance (QA) personnel, the plant engineer, the plant operator, and the production manager.

For new plants, a list of potential hot spots should be included in the hand-over documentation.

For new plant or product, it may be necessary to sample all hot spots [2,27–33].

5 SAMPLING METHODS FOR CLEANING VALIDATION

Various sampling methods used for cleaning validation are the following:

1. Swabbing (or direct surface sampling) method
2. Rinse sampling method
3. Placebo method
4. Microbial sampling.

5.1 Swabbing Technique

Swab sampling involves the use of a swabbing material, often saturated with solvent, to physically sample the surfaces.

Advantages

- Dissolves and physically removes a sample of the contaminant material.
- Adaptable to a wide variety of surfaces.
- Economical and widely available.
- May allow sampling of a defined area.
- Applicable to active, microbial, and cleaning agent residues.

Disadvantages

- An invasive technique that may introduce fibers.
- Results may be technique dependent.
- Swab material and design may inhibit recovery of the sampled material and specificity of the method.

- Evaluation of large, complex, and hard-to-reach areas difficult (e.g., crevices, pipes, valves, large vessels).
- Subject to vagaries of site selection.

The personnel performing the sampling operation need to be qualified. There should be training documentation to demonstrate the ability of the samplers to follow the sampling instructions.

The swabbing method needs to be very efficient, reproducible, and documented. A “random rub” method is usually not reproducible in swab sampling. The “squeegee method” is usually performed to meet swab sampling criteria. This method entails passing the swab (Fig. 5.3) over a defined area in two different directions, first from top to bottom and then from side to side in the specified surface area (Figs. 5.4 and 5.5).

5.2 Rinse Sampling

Rinse sampling involves passing a known volume of solution over a large area and analyzing the collected solution.

Advantages

- Adaptable to online monitoring.
- Easy to sample.
- Nonintrusive.
- Less technique dependent than swabs.





FIGURE 5.4 Swab sampling of equipment.

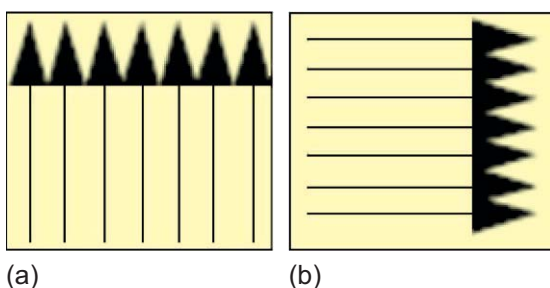


FIGURE 5.5 Squeegee method of swab sampling (a) from top to bottom and (b) from side to side.

- Applicable to actives, cleaning agents, and excipients.
- Allows sampling of a large surface area.
- Allows sampling of unique (e.g., porous) surfaces.

Disadvantages

- Limited information about actual surface cleanliness in some cases.
- May lower test sensitivity.
- Residues may not be homogeneously distributed.
- Inability to detect location of residues.
- Proper rinse volume is critical to ensure accurate interpretation of results.
- Sampling methodology must be defined since rinse sampling method and location can influence results.
- May be difficult to accurately define and control the areas sampled, therefore usually used for rinsing an entire piece of equipment, such as a vessel.
- Reduced physical sampling of the surface since surfaces of some parts cannot be accessed.

5.3 Placebo Sampling

Placebo sampling can be used to detect residues on equipment through processing of a placebo batch subsequent to the cleaning process. It is appropriate for active residues, cleaning agents, particulates, and microbial testing. Placebos are used primarily to demonstrate the lack of carryover to the next product. The placebo should mimic product attributes. The equipment characteristics also impact the choice of the placebo batch size.

Advantages

- Placebo contacts the same surfaces as the product.
- Applicable on hard-to-reach surfaces.
- Requires no additional sampling steps.

Disadvantages

- Difficult to determine recovery (contaminants may not be evenly distributed in the placebo).
- Lowers analytical specificity and inhibits detectability.
- Takes longer and adds expense since equipment must be cleaned after the placebo run.
- Placebos must be appropriate for each product.
- Residues may not be homogeneously distributed.
- No direct measurement of residues on product contact surfaces.

The preferred sampling method and the one considered as the most acceptable by regulatory authorities is the swabbing method.

5.4 Microbiological Sampling

To evaluate microbial bioburden direct contact plates {replicate organism detection and counting} provide quantitative analysis and are suitable for flat surfaces. Swabbing is suitable for irregular surfaces; it is more qualitative than quantitative; and it has been accepted as a satisfactory method due to its adaptability to a variety of surfaces.

6 CLEANING METHOD SPECIFICITY

A key parameter for cleaning validation methodology is to obtain maximum recovery from the selected solvent (or diluent) and swab combination. The cleaning process involves several factors, which include

- Selection of cleaning solvent
- Selection of diluent
- Selection of swab material
- Selection of extraction solvent.

6.1 Recovery in Swab Sampling

There are several areas of interest in recovery of materials in swab sampling.

6.1.1 Chemical Recovery from Spiked Swabs

The recovery from the spiked swab is determined separately from the spiked plate in order to give an indication of the efficiency of the solvent versus swabbing process. The recovery from a spiked swab should be 90% or greater to assure an acceptable recovery from the spiked plate. The recovery obtained from spiked swabs should be determined by spiking the swabs minimally at four concentrations across the range of approximately between 50% and 125% of the targeted concentration.

6.1.2 Recovery from Spiked Plates/Coupons

The recovery obtained from spiked plates should be determined similarly to the spiked swabs (Fig. 5.6). At least four concentrations across the range of approximately between 50% and 125% of the targeted concentration should be used. The recovery from a spiked plate should be greater than 75% to assure that an accurate representation of surface cleanliness is obtained.

6.1.3 Microbial Recovery from Spiked Swabs and Plates/Coupons

For microbial recovery, sufficient microorganisms should be seeded to allow recovery, which may vary depending on both the microorganism and the method used. Generally, recoveries using direct contact plates are greater than for swabbing or rinse methods.

6.2 Stability Issues in Cleaning Method

Several stability issues in the cleaning method need to be considered.



FIGURE 5.6 Recovery from spiked coupon.

6.2.1 *Surface Aging*

For chemical analysis, the length of time after cleaning of process equipment should be determined during which swab sampling must be performed.

6.2.2 *Swab Aging*

The length of time in which the swabs must be extracted should be determined.

6.2.3 *Timing of Sampling*

In general, after a surface has been cleaned, ensure that it is at least visually clean (i.e., residue free) and dry before sampling the sites. Do not swab any surface area until it has passed the appropriate qualitative (or visual) inspection acceptance criteria.

6.2.4 *Time Between End of Manufacturing to Beginning of Cleaning*

The duration of hold time between the end of manufacturing and the beginning of cleaning should be discussed and evaluated during the cleaning validation (this is because cleanability varies for different products). The data generated will help in the development of the standard operating procedure (SOP) during routine production condition.

6.2.5 *Number of Sample Sites*

Identify the sites with the most product buildup that are difficult to access when the equipment is not disassembled—sites which may have significant product flow. The number of sample sites to be selected should be based on the following:

1. Equipment and product contact surfaces
2. Adequate coverage to provide a high degree of confidence that the validated cleaning procedure ensures that the overall equipment has been properly cleaned per the cleaning SOP.

6.2.6 *Diagram of Sampling Sites*

Diagrams or still photos may be used to identify the sampling sites to help the individuals doing the sampling effectively.

6.2.7 *Microbiological Sampling Sites*

Factors to consider when choosing sampling sites include the following:

- High product contact locations where microbial contamination would most likely affect product quality.
- Sites most likely to support microbial growth and represent the most inaccessible or difficult areas to decontaminate (worst case).
- Areas to sample include valves, transfer lines, pumps, and gaskets.

6.2.8 *Sample Storage and Identification*

The sample should be stored in a suitable container to prevent contamination or alteration during storage. Microbiological samples should be protected from contamination and desiccation.

The storage container label includes the following content:

- Last batch/lot number
- Equipment number
- Last product name
- Sample number and description
- Date and time of sample collection
- Sample expiration time and date
- Sampler signature with date
- Comment section.

6.2.9 *Sampling Documentation: Data Sheet*

Documentation of the sampling during cleaning validation is necessary. It includes the following information:

- Sample date/time
- Sampled by (name)
- Sampling procedure/methodology
- Sampling site(s)
- Time/sample delivered date to test laboratory
- Special sample storage instructions
- Comments (to record any deviations)
- Past/present experiences [2,7,10,33–37,38–41].

7 SOLVENTS USED FOR CLEANING

Water is commonly used as a solvent for cleaning the equipment, but often other chemical solvents, such as 2-butoxyethanol (butyl), isopropyl alcohol (rubbing alcohol), and d-Limonene, are used to break up grease and oils to clean the equipment. Under some circumstances for cleaning in API industries, the same solvent is used for cleaning that is used during the synthesis due to solubility of the API. Commonly used solvents in API industry are methanol, acetone, dimethyl formamide, and ethyl acetate.

Advantages of organic solvent cleaning

- The API is usually soluble in the organic solvent.
- The solvent may be readily available and routinely used in the manufacturing process.

- Solvent residue analysis is simple and may be unnecessary if the cleaning solvent is the same as the process solvent used in the next batch.

Disadvantages of organic solvent cleaning

- Residues other than the active ingredient, such as degraded products and other byproducts, may be soluble in the cleaning solvent if they are present on the surface.
- The traditional approach of refluxing is time consuming.
- As the solvent evaporates, the residue also has the potential to redeposit on surfaces.
- Discarding large amounts of cleaning solvent can be an issue.
- Solvent can be recovered, but it adds to the overall cost of manufacturing.

7.1 Builders

An alternative to chelating agents are builders used to reduce the cost of the formulating the detergent, including phosphates and sodium carbonate.

Advantages of builders

- Added to the cleaning compound to upgrade and preserve cleaning efficiency of the surfactants.
- Have various functions like softening, buffering, and emulsifying.
- Builders soften water by deactivating hard minerals (metal ions like calcium and magnesium).
- Builders also provide a desirable level of alkalinity (increased pH) which aids in cleaning.
- Builders also act as buffers to maintain proper alkalinity in the wash water.

7.2 Cleaning Cycle

If the product residue in the equipment is buffer or salt, then it can be easily solubilized in hot water, and effective cleaning can be performed only using water rinsing. If the product residue is a biological compound, then the cleaning cycle can be performed with a series of washing processes like prewash, alkali, and acid cleaning, and final rinse of water for injection (WFI).

7.2.1 Prewash

- The prewash can be with hot or cold rinse, and it depends highly on type of residue.
- The prewash helps to get rid of some of the materials, such as residual sugar and salts, which are more soluble in purified water (PW) or WFI.

- If a protein is used, the prewash should be performed at ambient temperature, as elevated temperature will denature the protein and it will stick to the vessel surface which will be difficult to clean later.
- The wash water can be discharged directly into the drain without recirculation.

7.2.2 *Alkali Wash*

- The alkali is supplied by the feed pump until the set point reaches the supply tank.
- Detergent solution can be heated by passing through a heat exchanger.
- Recirculate for specified amount of time and then discharge to the drain.
- This step dissolves the residues which are not cleaned by the prewash.

7.2.3 *Postalkali Wash*

- All transfer lines and the vessel should be washed with hot WFI or PW.
- Alkali traces after the alkali wash can be cleaned.
- It can be used in recirculation mode or it can be discharged directly to the drain.
- The temperature in this step is raised with a heat exchanger supplied with steam.

7.2.4 *Acid Wash*

- Useful to remove specific residues which are not cleaned by alkali and WFI rinses, for example, protein residues that are more soluble in acids than in alkali solutions.
- The acidic wash can be performed with mild heating, as foam formation is observed at elevated temperature during acid washes.
- It can also be used for neutralization after the alkali wash.
- Recirculate the wash solution for a specific amount of period and then discharge to the drain.

7.2.5 *Final Rinse*

- Final rinse is performed until the conductivity of the final rinse solution equivalent to WFI is obtained, or it reaches a given set point.
- The temperature can be ambient, or at 70-80 °C to clean the remaining traces of acid.
- Final rinse can be once through discharge or it can be drained intermittently.

7.2.6 *Air Flushing for Storage*

- Air flushing can be used after each wash or after the final rinse only.
- It is used for removal of trace WFI or PW from the cleaned vessel and the transfer lines.

- It is used to dry the system.
- If possible, every cleaning cycle should end with this step for better cleaning.

8 CLEANING AGENTS

Key considerations in selection and use of cleaning agents are discussed in the following sections.

8.1 Grouping of Cleaning Agents

When selecting a new cleaning agent or utilizing an established cleaning agent for a new process, it is important to know all of the ingredients which are in the cleaning agent, along with the percentage of each constituent because removal of the cleaning agent must be verified in the cleaning process. Using a single cleaning agent greatly reduces the work required to determine if residues of the agent persist after cleaning. In multiproduct facilities, it may be necessary to use different cleaning agents to remove the various excipients that might be present in the products.

Generally, cleaning agents are detergents that are frequently used to remove the residues from surfaces. Detergents should be easily removable during the cleaning process. Simple water rinses may be adequate for the removal of highly soluble materials. Acceptable limits should also be defined for cleaning agent residues after cleaning. The possibility of cleaning agent breakdown should also be considered when validating cleaning procedures.

During the development of the cleaning process, the quantity of the cleaning agent, its concentration, and its addition rate, along with methods of storage, expiration date, inventory control, and change control of the cleaning agent, can help to establish and maintain a reproducible cleaning process. Once the ingredients are known, the company must determine the worst-case ingredient in the cleaning agent. In the pharmaceutical industry, cleaning agents are usually preferred that are water soluble by nature so that the removal of the cleaning agent is not a problem.

Two types of cleaning detergents are used: alkaline-based or acid-based detergents that are often formulated with surfactants, chelating agents, and emulsifiers to improve the efficiency of the detergents. [Table 5.4](#) compares the advantages and disadvantages of acidic and alkaline detergents.

The effectiveness of a cleaning agent when in contact with the residue that is to be removed may depend upon parameters such as cleaning agent concentration, exposure time, pH, temperature, and pressure. If the residue is thick and greasy, pressure may be increased to remove it completely from the surface of the equipment.

TABLE 5.4 Advantages and Disadvantages of Acidic and Alkaline Detergents

Acidic	Alkaline
● Inorganic soils are more soluble in acidic detergents ● Works well as blends of several acids	● Make organic soils water soluble via the change of chemical/physical nature of organic soils ● Easy to select (selecting an appropriate alkaline detergent is easy)
● Equipment corrosion ● Difficult to select (selecting an appropriate acidic detergent is difficult) ● Less compatible with other cleaning components	● Hard water will precipitate in this medium ● No cleaning effect on mineral residues ● Limited to organics ● Very concentration dependent

8.2 Selection of a Cleaning Agent

Depending on the particular type of chemical compound to be removed, the basis for selection of a cleaning agent includes the following:

- Chemical/physical nature of the compound
- Reactivity
- Physicochemical characteristics
- Chemical state of the compound.

Ideal characteristics of a cleaning agent are the following:

- Capable of wetting surfaces to penetrate solid deposits
- Have the capacity to break the solid into fine particles
- Ability to hold the small fine particles in suspension
- Prevent the residues from redepositing on the cleaned surface.

8.3 Water

Along with the cleaning agent, water is also an active ingredient that actually adds to detergency of cleaners. Water breaks up soil particles after the surfactants, reduces the surface tension, allows the water to penetrate the soil, and acts as a universal solvent. Water also keeps the soil suspended away from the clean surface, so that it can be maintained clean during the rinsing process, and it acts as an antiredeposition agent.

8.4 Alkaline Agents

Alkaline detergents are most effective in removing organic soils, such as oils, fats, proteins, starches, and carbohydrates, by hydrolyzing peptide bonds and

breaking down large, insoluble proteins into small, more easily soluble polypeptides in pharmaceutical and biopharmaceutical industries.

8.4.1 Sodium Hydroxide

Sodium hydroxide is widely used in breweries; it effectively dissolves protein substances and fatty oils by saponification. The ability of sodium hydroxide to dissolve proteins is enhanced by using it in combination with chlorine, surfactants, and chelating agents. It is very cheap and easily available, and it is easily soluble in water.

8.4.2 Sodium Hydroxide/Hypochlorite Solutions

Sodium hydroxide and hypochlorite mixtures are particularly effective in removing tannin substances from surfaces. These mixtures can be used in CIP systems for occasional purge treatments or to brighten stainless steel.

8.5 Acidic Agents

Acidic detergents are often used in a two-step sequential cleaning regime with alkaline detergents. Heavy soils, oils, and glucans are unaffected by acidic detergents. Acidic detergents are also used for the prevention or removal of water scale and aluminum oxide. Acidic detergents are more effective against bacteria than alkaline detergents.

8.5.1 Phosphoric Acid

Some of the residues which are not cleaned with an alkali rinse can be easily removed with an acid wash. Cleaning performance of acids is greatly enhanced by adding an acid-stable surfactant, which promotes penetration of surface deposits and assists in the process of rinsing at the end of the cleaning process. Phosphoric acid is useful for removing protein residues, and it also acts as a neutralizing agent after the alkali wash [2,9,42,43].

9 ANALYTICAL METHODS

Analytical methods should be specific and sensitive to detect residues or contaminants. With advances in analytical technology, residues from the manufacturing and cleaning processes can be detected at very low levels. If contaminants or residues are not detected at very low levels, it does not mean that there is no residual contaminant present after cleaning. It only means that levels of contaminant greater than the sensitivity or detection limit of the analytical method are not present in the sample.

Analytical method development and validation may be required at low levels of the active ingredient, cleaning agent, or cleaning solvent and possibly

excipients. Solubility, conductivity, and pH of the active constituent of the detergent or the solvent can be helpful in the design and development of the process and equipment for the production scale. Specific and nonspecific are the two analytical methods used widely to detect any compound. The choice of using a specific or a nonspecific method can be difficult.

Companies should challenge the analytical method, in combination with the sampling method(s), which is used to show that contaminants can be recovered from the equipment surface and at various levels, i.e., 50% recovery, 90% recovery, etc. If a drug active ingredient is highly toxic, a specific analytical method is always recommended. Chromatographic methods are preferred for cleaning validation studies because of their sensitivity, specificity, and ability to quantify.

The basic requirements of an analytical method are given below:

- The sensitivity of the method shall be appropriate to the calculated contamination limit.
- The method shall be practical and rapid, and use the existing instrumentation in the company as far as possible.

It is always desirable to select the simplest method that can be used to meet the desired goal.

The selected method should be practical, rapid, simple, and employ, as much as possible, the available instrument to meet the desired goal. The analytical development should include a recovery study to challenge the sampling and testing methods.

The analytical method should be validated as per regulatory requirements (Fig. 5.7). The steps involved in swab sampling are shown in Fig. 5.8.

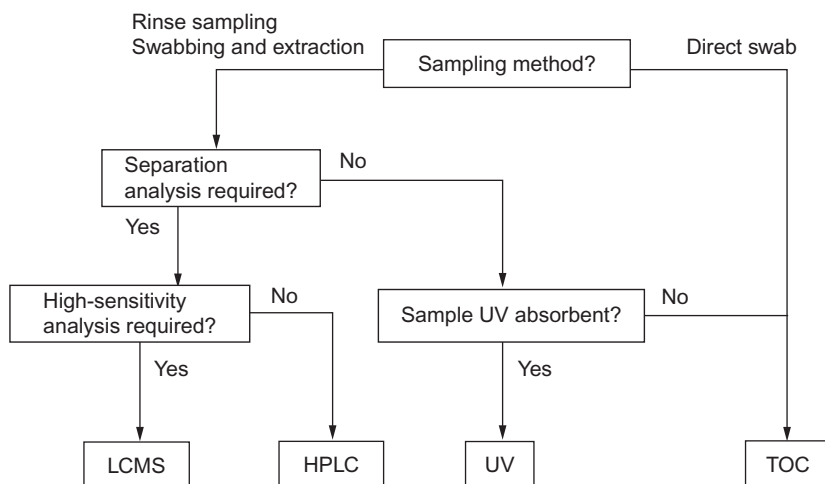


FIGURE 5.7 Decision flow to select an analytical method for cleaning validation.

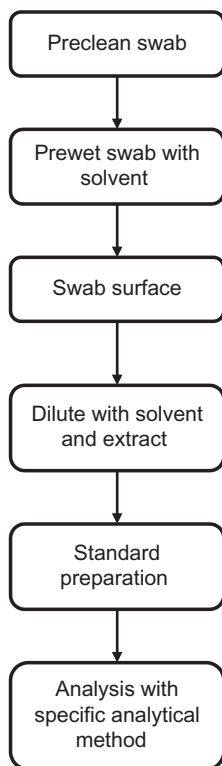


FIGURE 5.8 General steps for swab sampling and analysis.

9.1 Specific Methods

Specific methods detect a unique compound in the presence of potential contaminants. Specific methods include high-performance liquid chromatography (HPLC), ion chromatography, atomic absorption, capillary electrophoresis, and other chromatographic methods.

9.2 Nonspecific Methods

Nonspecific methods are methods that detect any compound that produces a certain response.

Nonspecific methods include total organic carbon (TOC), pH, titration, and conductivity.

For biological drug substances, product-specific assay(s), such as immunoassay(s) to monitor the presence of biological carryover, may not be acceptable; a negative test may be performed to ensure cleaning. Product-specific assay(s)

can be used in addition to total organic carbon (TOC) for the detection of protein residue [31,35,44–47].

9.3 Various Analytical Techniques in Cleaning Validation

There are a number of analytical techniques for cleaning validation. These are discussed below.

9.3.1 pH

The readings can be obtained directly by inserting a pH probe into the sample solution:

Advantages

- Rapid and inexpensive.
- Adaptable to online monitoring.

Disadvantages

- Nonspecific.
- Not appropriate for specific validation experiments.
- Applicable to water-soluble materials only [48,49].

9.3.2 Conductivity

Conductivity is strongly sensitive to the presence of any ionic or soluble inorganic contaminant. Conductivity can be measured by using a probe unit.

Advantages

- Rapid and inexpensive.
- Adaptable to online monitoring.

Disadvantages

- Nonspecific.
- Not appropriate for specific validation experiments [50,51].

9.3.3 Total Organic Carbon

The TOC method is based on oxidizing the organic compound and measuring the carbon dioxide produced. The residues detected are from potentially the most potent or toxic contaminant, which is usually the active ingredient.

Advantages

- Applies to a broad spectrum of contaminants.
- Low-level detection.
- Online capability.
- Rapid turnaround.

Disadvantages

- Nonspecific.
- Applicable to aqueous soluble samples only [52–56].



FIGURE 5.9 Luminescence of luciferin and its derivatives.

9.3.4 Enzymatic (Bioluminescence)

The chemistry is based on the reaction of ATP with luciferin/luciferase to generate bioluminescence which can be quantitated with a simple, portable instrument (Fig. 5.9).

Advantages

- Highly specific.
- Very sensitive.

Disadvantages

- Expensive.
- Difficult to develop and validate.
- May not provide accurate results if proteins are denatured [57,58].

9.3.5 Light Microscopy

Microscopy is a method of identifying contaminants on equipment and within products. Microscopic techniques employed include light microscopy and scanning electron microscopy, as well as light microscopy with X-ray diffraction, mass spectrometry, and nuclear magnetic resonance.

Advantages

- Provides quick qualitative identification of contaminants.
- Can complement results from nonspecific quantitative techniques such as gravimetric analysis and TOC.

Disadvantages

- Not quantitative.
- Somewhat subjective [2,59–63].

9.3.6 Gravimetric Method

The gravimetric method involves rinsing the entire equipment, evaporating the rinse solvent to dryness, and weighing the residue.

Advantages

- Applies to a broad spectrum of contaminants.
- Simple and low cost.

Disadvantage

- Nonspecific [2].

9.3.7 Thin-Layer Chromatography

Thin-layer chromatography has been used for the analysis of residues of actives and cleaning agents.

Advantages

- Highly specific.
- Moderate-to-high sensitivity.
- Fairly inexpensive.

Disadvantages

- Visual endpoint detection is not quantitative.
- Automatic readers are semiquantitative.
- Lengthy process to perform sample preparation [64].

9.3.8 Capillary Zone Electrophoresis

Also known as capillary electrophoresis, this technique has been applied mostly in the biotechnology industry and is effective for evaluating residues of proteins, amino acids, and certain cleaning agents. [Figure 5.10](#) shows a schematic of the technique.

Advantages

- Highly specific.
- Highly quantitative.
- Sensitive.

Disadvantage

- Expensive [2,65].

9.3.9 Fourier Transform Infrared Spectroscopy

Fourier transform infrared spectroscopy involves the application of advanced mathematical concepts to generate Fourier transforms of multiple infrared scans of a sample.

Advantages

- Specific.
- Qualitative.
- Can be quantitative.

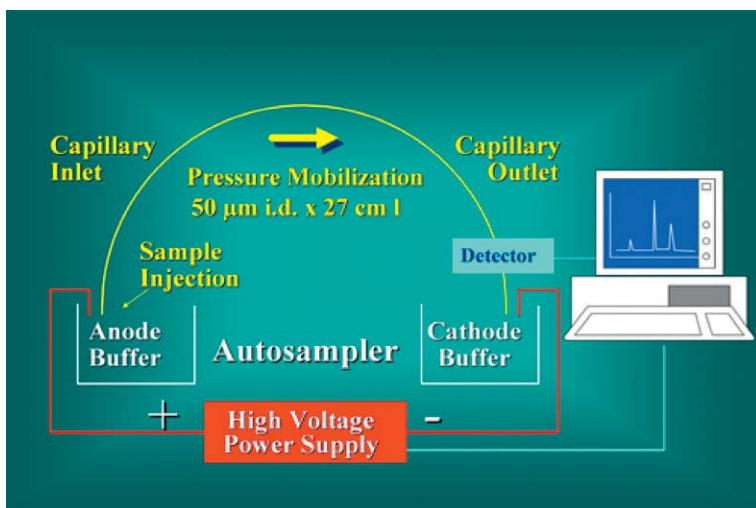


FIGURE 5.10 Schematic of the capillary zone electrophoresis technique.

Disadvantages

- Expensive.
- Requires an extensive library of spectra [66,67].

9.3.10 Enzyme-Linked Immunosorbent Assay

An enzyme-linked immunosorbent assay is an antigen-antibody type reaction involving the use of specific chemicals.

Advantages

- Highly specific.
- Very sensitive.

Disadvantages

- Very expensive.
- Difficult to develop and validate.
- Labor intensive.
- May not provide accurate results if proteins are denatured [2,68].

9.3.11 Atomic Absorption Spectroscopy

It deals with the absorption of specific wavelength of radiation by neutral atoms in the ground state.

Advantages

- Very specific.
- Sensitive.

Disadvantages

- Generally useful only for metals, salts, and metal complexes.
- Expensive [69–71].

9.3.12 Ultraviolet Spectrophotometry

It deals with the measurement and interpretation of electromagnetic radiation absorbed or emitted when the molecules or atoms or ions of the sample transition from one energy state to another energy state.

Advantages

- Moderately to highly specific.
- High sensitivity.
- May be used as a screening method or for confirmatory identification.

Disadvantage

- Requires more technical expertise and more expensive equipment than some of the other methods [72,73].

9.3.13 Microbial and Endotoxin Detection and Testing

Microbial testing method is based on isolating, quantitating, and speciating bacteria by microbiological techniques. Sterile swabs and samples from rinse solutions are used as vehicles to generate samples for microbial testing and endotoxins can be detected by gel clot, chromogenic, and turbidimetric limulus amebocyte lysate methods or by rabbit pyrogen methods [2].

10 CLEANING DEVELOPMENT PHASE

The SOPs and operator training should be developed in the cleaning development phase.

10.1 Standard Operating Procedures

SOPs should be developed in conjunction with the cleaning development phase. After the cleaning process has been validated, final SOPs for cleaning should be highly detailed.

The following items should be validated, documented, and included in the appropriate SOP:

- Extent of disassembly of the equipment. Disassembly should be such that the equipment is broken down in a manner that will allow all parts to be effectively cleaned.
- Visual inspection for equipment wear, excessive product residuals, and foreign materials.
- Critical sites or difficult to clean areas that may require special cleaning emphasis or a specific inspection.

- Assignment of responsibility for cleaning equipment.
- Cleaning schedules where appropriate and sanitizing schedules.
- A description in sufficient detail of the methods, equipment, and materials to be used in cleaning, and the methods of disassembling and reassembling equipment as necessary to ensure proper cleaning as well as the cleaning/handling and storage of tools used in cleaning (postcleaning).
- Removal or obliteration of identification information of the previous batch.
- Protection of clean equipment from contamination prior to use.
- Inspection of equipment for cleanliness immediately before use [2,42,74,75].

10.2 Operator

All personnel must be trained and each operator must understand the cleaning steps and process.

It is important that not only operator training occurs, but also the training should be well documented. Without proper documentation, it is impossible to prove that the training was actually accomplished. Operators should be retrained whenever there is any change in cleaning procedure and new training documentation must be created.

10.3 Operator Training

Operator training is critical, especially for manual cleaning. During the development of the cleaning process cycle, operators should be trained in the existing SOPs. Proper training consists of understanding the SOP, apprenticeship with qualified, trained operators, and review to ensure that the operators successfully completed their training. The effective training or qualification of the operators may be confirmed by monitoring equipment cleaning, including where necessary, analytical testing for residues from the cleaned equipment.

Training practices may vary from one organization to another, but operator training must be enhanced by

- Clearly written, understandable, and sufficiently detailed SOPs;
- Use of checklists to determine that all operations are carried out in the proper sequence and are documented;
- Periodic monitoring of cleaning processes to ensure proper training of operators and continued compliance with SOPs;
- Dedicated or assigned cleaning personnel;
- Feedback from operators to modify procedures;
- Use of video to demonstrate proper cleaning operations and techniques.

Overall, the operators should understand the process of cleaning and the operation of the equipment they are cleaning. In addition, the operators should be aware of the impact of the cleaning process on the quality of the next product manufactured in the same equipment [2,9,10,75,76].

11 CLEANING VALIDATION PROTOCOL

The cleaning validation protocol should be formally approved by the plant management to ensure that aspects relating to the work are defined in the protocol. A cleaning validation protocol is required that lays out the procedure for how the cleaning process will be validated. It should include the following information:

- The objective of the validation process.
- Responsibilities for performing and approving the validation study.
- Description of the equipment to be used.
- The interval between the end of production and the beginning of the cleaning procedures.
- Cleaning procedures to be used for each product, each manufacturing system, or each piece of equipment.
- The number of cleaning cycles to be performed consecutively.
- Any routine monitoring requirement.
- Sampling procedures, including the rationale for why a certain sampling method is used.
- Clearly defined sampling locations.
- Data on recovery studies where appropriate.
- Analytical methods including the limit of detection and the limit of quantitation of those methods.
- The acceptance criteria, including the rationale for setting the specific limits.
- Other products, processes, and equipment for which the planned validation is valid according to “bracketing” concept.

Records should be maintained about the cleaning performed with the following information:

- The area or piece of equipment cleaned.
- The person who carried out the cleaning.
- When the cleaning was carried out.
- The SOP defining the cleaning process.
- The product which was previously processed on the equipment being cleaned.

The cleaning record should be signed by the operator who performed the cleaning, followed by the person responsible for production, and finally the record should be reviewed by the QA department [9,10,77,78].

11.1 A Model Cleaning Validation Protocol

This section describes the key features of a model cleaning validation protocol.

1. *Preapproval protocol*

Signing of this preapproval protocol indicates agreement with the cleaning validation approach established for the particular location; further, if any changes in this master plan would be required, it will be revised and duly approved by the validation team members (quality control, production, and QA).

2. *Introduction*

Cleaning validation in the context of manufacture of the products at <COMPANY NAME> may be defined as

“The process of providing documented evidence that the cleaning method of the equipments and ancillary utensils employed within the facility consistently controls potential carryover of product (including intermediates and impurities), cleaning agents and extraneous material into subsequent product to a level which is below predetermined levels.”

3. *Objective*

The objective of this protocol is

- To establish and assure with documented evidence that the cleaning procedures used after manufacturing of <PRODUCT NAME> are effective and consistently perform as expected and produce a result that meets a predetermined acceptance criterion when manufactured.
- To provide documented evidence through the scientific data to show that the cleaning procedure used after manufacturing of <PRODUCT NAME> is effective and consistently performs as expected and produces a result that meets predetermined acceptance criteria when manufactured.
- To establish that the cleaning process shall provide a high degree of assurance for removal of residues of the last manufactured product so that those residues are not transferred to the subsequently manufactured product.
- To prove that equipment is consistently cleaned of product to an acceptable level to prevent contamination and cross-contamination.

4. *Scope*

The scope of this protocol is applicable for cleaning validation of <NAME OF THE PRODUCT WITH API STRENGTH>. It is also to evaluate the acceptability of cleaning procedure used in cleaning of equipment using well-established analytical and microbiological methods to determine the chemical and microbiological residue after cleaning of the equipment. This protocol will also cover the responsibilities, sampling plan, acceptance criteria, revalidation criteria, and change control procedure. This protocol is applicable for <NAME OF THE COMPANY>.

5. *Responsibilities*

Quality Assurance

- To prepare the cleaning validation protocol
- To review the cleaning validation protocol

- To monitor cleaning activities
- To collect samples from cleaned equipment
- To review analytical reports
- To prepare and review the cleaning validation report.

Production

- To review the cleaning validation protocol
- To provide details of equipments
- To have the equipment cleaned by trained operators
- To schedule cleaning validation program
- To maintain log books for equipment cleaning
- To review the cleaning validation report.

Quality Control

- To review the cleaning validation protocol
- To analyze the samples collected for cleaning validation
- To compile analytical data
- To review the cleaning validation report.

Quality Control (Microbiology)

- To review the cleaning validation protocol
- To collect microbiological samples from cleaned equipment
- To analyze the cleaning validation samples
- To compile the microbiological data
- To review the cleaning validation report.

Head of QA

- To review and to approve the cleaning validation protocol and report.

6. Protocol Training Record

- All the personnel involved in the cleaning validation activity, sampling, and testing of cleaning validation samples must be appropriately trained in their assigned job responsibilities and on the cleaning validation protocol.
- Personnel or operator who performs cleaning routinely should be trained and should be effectively supervised.
- Record the training details of the persons involved in sampling and testing of the product.

7. Product Profile

- <NAME OF THE API WITH STRENGTH> is the existing product of the <LOCATION, WHERE THE VALIDATION IS TO BE PERFORMED> of <COMPANY NAME>.
- <A STATEMENT RELATED TO THE SOLUBILITY PROFILE OF THE API>.
- <IUPAC NAME OF THE API>.
- <CAS NUMBER OF API>.
- Chemical Skeletal Structure of the API.
- <A STATEMENT RELATED TO THE POTENCY OF THE API>.

- <A STATEMENT RELATED TO THE EQUIPMENT USAGE BY THE API>.
- Three batches of the product shall be validated for cleaning.

8. *Equipment Description*

Following equipment of same design and operating principles shall be deployed for cleaning validation of the three batches of <NAME OF PRODUCT>:

S. No.	Name of Equipment	Capacity	ID Number	Surface Area (cm ²)	**Surface Area (cm ²)
1	<EQUIPMENT 1>				
2	<EQUIPMENT 2>				
3	<EQUIPMENT 3>				
4	<EQUIPMENT 4>				

** Surface Area + 10% of Surface Area <This neutralizes any calculation errors and also takes ancillary equipment like scoops, spatulas, etc. into account>.

9. *Methods of Cleaning*

Information about the cleaning methods is included here:

9.1. There are three types of cleaning procedures deployed at <COMPANY NAME>

<WE ARE TAKING THE FOLLOWING CLEANING TYPES AS EXAMPLE>

(a) **Type A**

- For batch to batch changeover.
- End of the shift cleaning or whenever required.
- Changeover to product of ascending strength, but of same color and flavor.

(b) **Type B**

- During changeover of product with different APIs, color/ flavor and products having same API/color/flavor but with descending strengths.
- After five consecutive batches of the same product.
- Equipment kept in idle condition for more than 48 h.
- Equipment kept in idle condition for more than 7 days subsequent to Type “B” cleaning.
- After carrying out preventive maintenance or any major maintenance activity on the equipment.

(c) **Type C**

- On “Type B” cleaned equipment just prior to use, when required to be used within 7 days from the date of Type “B” cleaning.

9.2 **Materials and Equipment Used for Cleaning**

- | | |
|--------------------|-----------------------------------|
| (a) Purified water | (d) Scrubber |
| (b) Tap water | (e) Dry sponge or lint-free cloth |
| (c) Vacuum cleaner | (f) <NAME OF THE DETERGENT> |

9.3 Cleaning SOP for Each Equipment Part

S. No.	Name of Equipment	ID Number	Cleaning SOP Number
1	<EQUIPMENT 1>		
2	<EQUIPMENT 2>		
3	<EQUIPMENT 3>		
4	<EQUIPMENT 4>		

10. Sampling Method

Information about the sampling method is included in this section.

10.1. Selection of Sampling Method

Swab sampling shall be considered as the sampling method.

Justification for swab sampling: Looking at the design and size of the equipment, swab sampling shall be considered the main method for validation; however, rinses will also be obtained wherever necessary. Most difficult to clean locations are selected for sampling to determine the efficacy of cleaning.

Advantages of swab sampling:

- Direct evaluation of surface contamination.
- Insoluble and poorly soluble substances may be physically removed.
- Hard to clean but accessible areas are easily incorporated in the final result.

10.2. Scientific Rationale for Selection of Sampling Points

The product contact surface area which is most difficult to clean shall be selected as the sampling point. The locations selected for swabbing are generally those locations that are most difficult to clean, representative of different materials, and representative of different functional locations (side corners, agitator, blades, etc.). If these locations are swabbed and if residues in these materials are acceptable, then residues on another location shall also be acceptable. Performing swabbing in these locations and materials can be helpful in terms of providing higher assurance in the validation results.

10.3. Visual Inspection

After cleaning of the equipment, visual inspection shall be performed. To carry out a visual inspection, use a flashlight if required, and a mirror (attached to a stainless steel rod) to inspect the surface of equipment. This should be done under viewing conditions (lighting, angle, distance) that simulate viewing of the equipment.

10.4. Swab Sampling for Chemical Analysis of API

Description of swab:

- Make
- Model
- Swab tip material
- Swab stick material.

Swab samples shall be taken after final cleaning of the equipment, once the equipment passes the visual inspection test. The swab shall be dipped in the swabbing medium <SOLVENT FOR THE API AS PER THE METHOD VALIDATION> in a 50-mL test tube. Swab samples from different areas of the equipment shall be collected. The swab area shall be measured for swabbing.

Sampling error: During swab sampling, the following care should be taken: area of sampling should not be less than 100 cm²; apply proper force during collection of sample to avoid any sampling errors.

Sampling area: 10 cm × 10 cm = 100 cm² or equivalent (for the parts, where 100 cm² is not available as a contiguous whole area).

10.5. *Swabbing Pattern*

Apply the swab only once on a surface a single time only. Collect the swab by application of normal force. Note: Avoid lifting the swab stick at the contact surface during collection of the swab sample. Refer to the sampling diagram (Fig. 5.4) for sample collection using a swab.

10.6. *Swab Sampling for Microbial Analysis*

- Sterile hand gloves and face mask shall be worn before removing the swab from the dispenser.
- A sterile swab shall be removed from the test tube and dipped into 0.9% sterile saline solution.
- The swab shall be stroked over a 5 cm × 5 cm (or equivalent) of the product contact surface.
- The strokes should be as per the procedure for chemical swabbing only.

Immediately after sample collection, the swab shall be processed per the procedure for microbial testing

10.7 *Swab Sampling Location*

Prior to swab sampling, cleaned equipment shall meet “visual clean” criterion. Sampling shall be carried out per the current version of the SOP. Chemical and microbial sampling locations shall be different from each other. The locations from where the swab sample is to be taken are identified below:

Chemical Swab Sampling			Microbial Swab Sampling	
S. No.	Equipment	Location	No. of Swab Locations	No. of Swab Locations
1	<EQUIPMENT 1>			
2	<EQUIPMENT 2>			
3	<EQUIPMENT 3>			

11. Analytical Procedure

Analytical methods for specific analyses shall be validated per the analytical method validation protocol. Validated methods shall be employed to analyze the cleaning validation samples.

12. Establishment of Acceptance Criteria

The cleaning procedure shall be considered validated, when the acceptance criteria, as specified in the protocol, are met. Failure of individual sampling points will not necessarily mean that the cleaning method is inadequate. Each deviation should be investigated and, based on the investigation, corrective actions will be taken with further follow-up or further validation.

12.1 Visual Inspection

Equipment should be visually clean and dry and must contain NO visible residues.

12.2 Active Residue

Calculation of active residue after cleaning shall be based on product contact surface area. This approach is based on acceptable daily intake. Based on the acceptable daily intake and safety factor (1000 for oral dosage forms), the maximum allowable carry over (MACO) is calculated. It is assumed that only a fraction (1/1000) of the smallest daily dose of product A can be carried over to the maximum allowable daily dose of product B manufactured in the same equipment train.

12.3 Rationale to Calculate Maximum Allowable Carryover Assuming the Worst Case

The maximum allowable carryover shall be calculated using the following items:

STD _(A)	Single Therapeutic Dose of Product A (in mg)
SBS _(B)	Smallest Batch Size of Product B (in mg)
SF	Safety Factor (constant) = 1000 for solid dosage forms
LDD _(B)	Largest Daily Dose of Product B (in mg)
MDD _(B)	Maximum Normal Daily Dose for Next Product

Thus, the formula to calculate MACO is:

$$\text{MACO for a specific equipment train} : \frac{\text{STD}_{(A)} \times \text{SBS}_{(B)}}{\text{SF} \times \text{MDD}_{(B)}}$$

12.4 Rationale to Calculate Maximum Allowable Carryover with the 10 ppm Criterion

The following approach can also be used to calculate MACO for the products manufactured in the tablet section. The starting point is the amount of contaminant (Product A) accepted as being carried over with the “next” Product B. The approach is to regard the active

ingredient of Product A as the contaminant to look for. For solids, an active ingredient intake of 1/1000th of the lowest therapeutic dosage of that active ingredient is usually regarded as harmless. MACO is calculated as follows.

- The maximum amount of contaminant Product A allowed to be taken in per day is $STD_{(A)}/1000$ per day. This means that with the daily intake of the next product, the maximum allowed contamination is 1/1000th of the daily therapeutic dosage of the contaminant, which has been mentioned in the master plan as “Safety Factor.” To be on the safe side, the maximum daily intake of the next product is taken into consideration.
- In case of 10 ppm (or 10 mg/kg) criterion, $STD_{(A)}$ is considered to be 10 mg/kg, regardless of the actual amount of dose.
- Thus, the quantity Q of Product A allowed in a single dose of Product B is calculated as:
 - $Q = 10/[1000 \times MDD_B \text{ (in numbers)}]$ or $0.01/MDD_B \text{ (in numbers)}$
 - This quantity is for one dose unit of Product B.
- For the entire batch of Product B, MACO is calculated as:
 - $MACO_{(10ppm)} = Q \times SBS_{(B)}/\text{One Dose Unit of Product B}$ (subject to recovery factor)
 - Here $SBS_{(B)}$ is the smallest batch size of product B.

12.5 Establishment of Acceptance Limits

The acceptance limits are established as described below.

12.5.1. Acceptance Limits per Equipment (ALE) for API

After calculating the MACO value for the whole equipment train, an acceptance limit of maximum allowable carryover per equipment is required to be calculated for individual equipments of the train. The ALE is calculated from the following expression:

$$ALE = \frac{MACO \times \text{surface area of individual equipment}}{\text{Total surface area of all equipment in the train}}$$

12.5.2. Acceptance Limits per Swab for API

Once the ALE value is obtained, the acceptance limit per swab (ALS) is calculated to get an acceptable quantity of maximum allowable carryover per swab. ALS is calculated from the expression below:

$$ALS = \frac{ALE \times \text{swab surface area}}{\text{Surface area of individual equipment}}$$

ALS for Chemical Analysis = <MENTION THE OBTAINED LIMIT>

Note: The swab surface area is considered as 10 cm × 10 cm or equivalent.

12.6 Acceptance limit for Microbial Bioburden

Swab samples for microbial analysis shall be collected from product contact surface area immediately after the completion of cleaning activities and after a specified hold time period of total aerobic microbial count until further usage of the equipment for manufacturing the next batch/product. The limits for microbiological bioburden criterion for product contact surface area are as follows:

Product Equipment Contact Surface	Microbiological Bioburden (cfu/25 cm ²)	Corrective Action (If the Counts Exceed the Limit)	
	Total Plate Count	Mold and Yeast	
Alert level	<AS PER THE REQUIREMENTS>	Absent	● No action required
Action level	<AS PER THE REQUIREMENTS>	Absent	● Investigate possible causes ● Perform recleaning ● Perform extra microbial testing
Limit	<AS PER THE REQUIREMENTS>	Absent	

13. Acceptance Limit for Cleaning Agent

To establish the effectiveness of cleaning process to remove the cleaning agent, <NAME OF THE DETERGENT>, the acceptance limits shall be prepared based on the toxicological data of the <DETERGENT>. The calculations are as follows:

$$\text{NOEL} = \text{LD}_{50}(\text{of Detergent}) \times 70\text{kg}/2000$$

$$\text{MACO}_D = \text{NOEL}(\text{of Detergent}) \times \text{SBS}/\text{SF} \times \text{MDD}_B$$

Here MACO_D is the maximum allowable carryover for the detergent; NOEL is no observed effect level; 2000 is an empirical constant; LBS_B is the largest batch size of Product B; MDD_B is the maximum normal daily dose for the next product; and SF is the safety factor, 1000 (constant for oral solid dosage).

Thus, the $\text{ALS} = \text{< CALCULATE AND DEFINE THE LIMIT >}$

14. Hold Time Study

To establish the effectiveness of cleaning, the equipment shall be kept idle for 72 h under controlled conditions. To establish the expiration of cleaning with respect to microbiological contamination, the equipment shall be kept idle after cleaning for 72 h and microbiological swab shall be taken and analyzed. This can be considered as the worst case and the microbial load shall remain within the limits.

15. Revalidation Criteria

The cleaning procedure needs to be validated only once. Periodic monitoring using swab samples is required to ensure compliance. Revalidation shall be performed when there is

- Change in the cleaning procedure;
- Change in cleaning agent used for cleaning;
- Change in minimum batch size and lowest dose of the product, i.e., change in MACO limit;
- Major modification in processing equipment;
- Any regulatory requirement(s).

Revision No.	Date of Review	Changes	Reviewed By
R0		First issue	

12 VALIDATION REPORT

A validation report is necessary to present the results and conclusions and secure approval of the study. The report should include the following information:

1. References to all the procedures followed to clean the samples and tests performed.
2. Physical and analytical test results or references for the same, as well as any pertinent observations.
3. Conclusions regarding the acceptability of the results, and the status of the procedures being validated.
4. Any recommendations based on the results or relevant information obtained during the study, including revalidation practices if applicable.
5. Review of any deviations from the protocol.
6. When it is unlikely that further batches of the product will be manufactured for a period of time, it is advisable to generate reports on a batch-by-batch basis until the product is again manufactured.
7. An appropriate cleaning validation report is to be prepared from the results of the cleaning validation study.

13 THE FDA CLEANING VALIDATION GUIDELINE

The FDA does not set specific acceptance limits for residue levels due to the great variety of equipment, processes, and products in the pharmaceutical industry. A company's rationale for the residue limits established should be logical based on the manufacturer's knowledge of the materials involved and be practical, achievable, and verifiable.

FDA expects the company to perform the cleaning validation three (3) times, according to FDA "if it comes out right once it is an accident, twice coincident, three times validation."

13.1 FDA Requirements

1. FDA expects companies to have written SOPs detailing the cleaning process used for various pieces of equipment.
2. If companies have a specific cleaning process for cleaning between different batches of the same product and use a different process for cleaning between product changes, FDA expects the written procedures to address these different scenarios.
3. If companies have one process for removing water-soluble residues and another process for water-insoluble residues, the written procedure should address both scenarios and make it clear when a given procedure is followed.
4. It is required by the FDA, in the general validation procedure, that the personnel responsible for performing and approving the study should comply with the acceptance criteria and the revalidation data.
5. FDA expects companies to prepare specific written validation protocols in advance for the studies to be performed on each manufacturing system or piece of equipment, which should address such issues as sampling procedures, and analytical methods to be used including the sensitivity of these methods.
6. It is expected that companies conduct the validation studies in accordance with the protocols and document the result of studies.
7. The final validation report is to be approved by the regulatory board which states whether or not the cleaning process is valid.

13.2 Acceptance Criteria

1. *Qualitative*

Visual inspection after cleaning must be performed and documented. The requirement for this component applies to all product contact surfaces of any equipment that has been cleaned and dried. The requirement is that the surfaces of the equipment must be inspected, to the extent possible, to verify that they are clean and free of any visible residue or film.

2. *Quantitative Analysis*

Must be met on a minimum of three separate, consecutive executions of a cleaning validation protocol for the cleaning process to be considered validated.

The requirement is composed of two elements:

- A. The average active drug residual test results of all samples for each major piece of equipment must be
 - Standard therapeutic dosage drug actives: No more than 100 µg per 100 cm².
 - Low therapeutic dosage drug actives: No more than 10 µg per 100 cm².

- B. The individual active drug residual test result must be
 - Standard therapeutic dosage drug actives: No more than 200 μg per 100 cm^2 .
 - Low therapeutic dosage drug actives: No more than 20 μg per 100 cm^2 .
- 3. *Analytically Clean for Cleaning Agent*
 The following two requirements apply to all product contact surfaces that have been cleaned and dried:
 - A. The average cleaner residual test results of all samples for each major piece of equipment must be
 - Standard LD₅₀ cleaning agents: No more than 200 μg per 100 cm^2 .
 - Low LD₅₀ cleaning agents: No more than 100 μg per 100 cm^2 .
 - B. The individual cleaner residual test result must be
 - Standard LD₅₀ cleaning agents: No more than 400 μg per 100 cm^2 .
 - Low LD₅₀ cleaning agents: No more than 200 μg per 100 cm^2 .

14 EFFECTIVE CLEANING VALIDATION MAINTENANCE PROGRAM

Maintaining an effective cleaning validation program is important to API manufacturing [10,42,78–82]. Written procedures shall be established and followed for cleaning and maintenance of equipment including utensils used in the manufacture, processing, packing, or holding of a drug product. These procedures include:

1. Assignment of responsibility for cleaning;
2. Maintenance and cleaning schedules;
3. A description of the methods, equipments, and materials used in cleaning and maintenance operations;
4. Records shall be kept of maintenance, cleaning, sanitizing, and inspection.

14.1 Equipment Cleaning Validation and Maintenance

When a minimum of three cleaning validation runs are completed and if the results meet the acceptance criteria, then the cleaning procedures would be demonstrated sufficiently and consistently to remove chemical and detergent residues from equipment surfaces during the study in order to meet the preestablished criteria. However, certain other factors can decrease the efficiency and consistency of the cleaning program over time. These factors are:

1. Operator variability.
2. Equipment aging and repair.
3. Potential nonrepresentative results and monitoring programs.
4. Changes to the product, equipment, and process.

14.2 Overview of Cleaning Validation Program

An overview of the cleaning validation program is shown in Fig. 5.11.

14.3 Cleaning Validation Lifecycle Management

Figure 5.12 shows the management of cleaning validation life cycle.

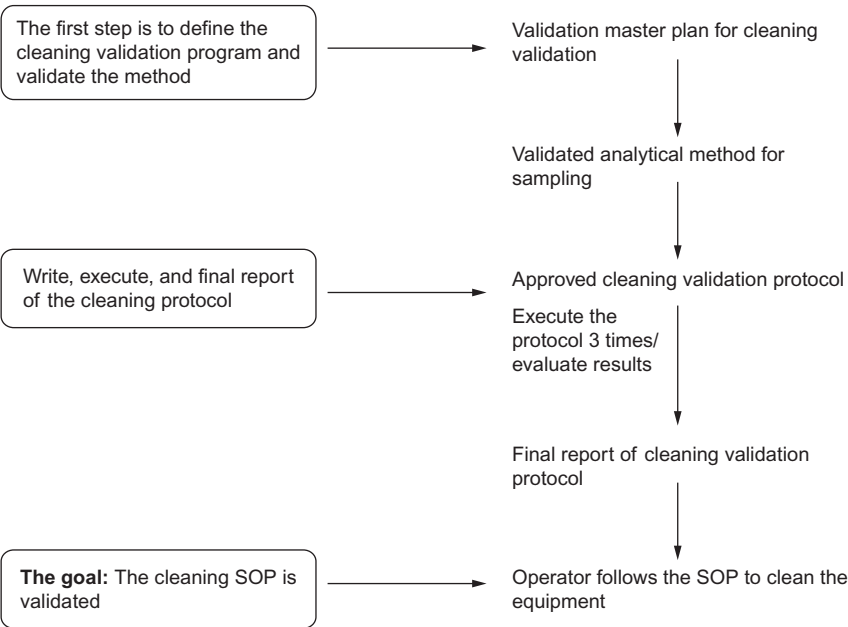
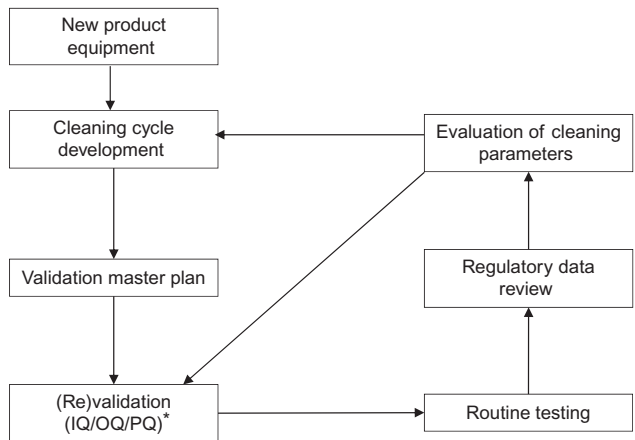


FIGURE 5.11 Overview of a cleaning validation program.



*IQ = installation qualification; OQ = operational qualification; PQ = performance qualification

FIGURE 5.12 Management of the cleaning validation life cycle.

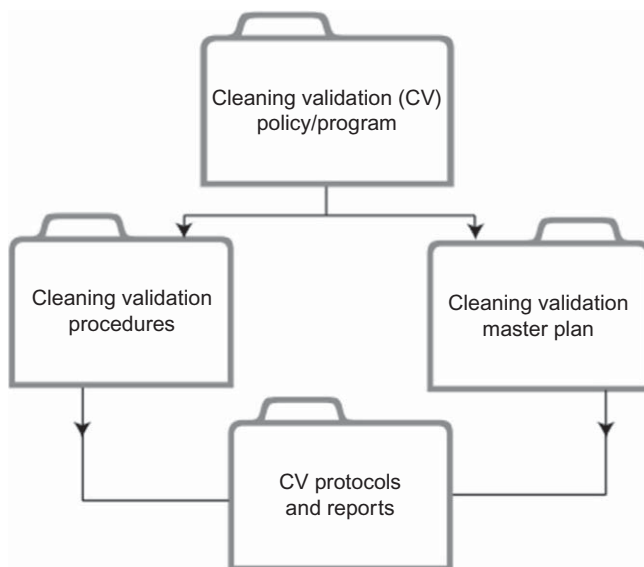


FIGURE 5.13 Overall cleaning validation process.

14.4 Cleaning Validation Chart

Figure 5.13 is a chart depicting the overall cleaning validation process.

15 SUMMARY

Cleaning validation is the most critical consideration in pharmaceutical industries. Inadequate cleaning can result in contamination of drug products with bacteria, endotoxins, active pharmaceuticals from previous batch runs, and cleaning solution residues. Such contaminants must be reduced to safe levels, both for regulatory approval and to ensure patient safety. Regulatory scrutiny is more rigorous in a multiproduct facility compared to a single product establishment. A regulatory expectation in cleaning validation is to ensure that residues from one product will not carry over and cross-contaminate the next product. Developing effective cleaning processes is a major part in establishment of API manufacturing, which takes a significant amount of time due to the complexity of API plant/equipment and the materials used in manufacture.

Companies are usually cited either for not having sound cleaning validation or meeting the protocol acceptance criteria. Failing a protocol acceptance criterion is considered a substantial regulatory risk; hence, companies are forced to spend money and resources even though there is minimal or no product risk. It is vital for successful cleaning validation to have appropriate acceptance criteria. In developing the acceptance criteria, companies may adopt a conservative approach either to prove that they have a sound cleaning validation program or to ensure that field data (results) will reflect the acceptance criteria.

Any number of approaches may be taken for different cleaning scenarios, different approaches may be more or less appropriate. Regardless of the approach taken, documentation of the rationale for the approach in establishing the acceptance criteria is necessary. The established acceptance residue limits must be logical, practical, achievable, and verifiable. The chosen analytical method in cleaning validation sample analysis should be familiar, exhibit robustness, ease of use, and regulatory acceptability.

Pharmaceutical plants must have visually clean equipment to operate according to good manufacturing practices. Formulators must visually inspect manufacturing equipment for cleanliness before formulation work begins.

In conclusion, the cleaning validation program should be based on detailed cleaning procedures, a good training program, a cleaning validation protocol, validated chemical and microbiological methods, a change control program, a final report, and any audit required to ensure compliance of the product.

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